Lukas Pfalz

Seeking Summer 2027 Robotics & Autonomy Internship | U.S. Citizen

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GitHub

Education

University of Maryland Global Campus — Remote
B.S. Computer Science | ***GPA 4.0*** | *Dean's List* — *Expected Dec 2027*

University of Maryland, College Park — College Park, MD
Coursework in Computer Science & Information Science — *Aug 2021 - Dec 2025*
Key courses: Data Structures & Algorithms, Linear Algebra, Calculus I-III, Probability & Statistics, Computer Architecture

Experience

KBR — Lexington Park, MD
Data Science Intern — *Jun 2026 - Present*

- Built an offline vision-language triage system for a ROS 2 Humble rover on a Jetson Orin NX: it approaches a person, interviews them by voice, and returns a severity-flagged assessment with escalation.
- Deployed YOLO26 on TensorRT (FP16) for posture detection and single-target tracking; a 20 Hz PID approach loop turns box center, depth range, and IMU gyro feedback into Ackermann steering.
- Fed detections as structured text to a local MedPsy-4B LLM alongside a live Whisper transcript; ran the wake-word STT/LLM/TTS loop on-device with Q4 GGUF quantization and sm_87 llama.cpp builds.

Forterra — Clarksburg, MD
Yard Software Intern — *May 2025 - Present*

- Built a Dockerized Python ETL pipeline that streams autonomous-fleet telemetry into PostgreSQL with idempotent upserts and fault-tolerant batching.
- Created Grafana dashboards tracking fleet uptime, autonomous vs. manual miles, and mission success.

Projects

SO-101 Robot Learning (in progress) — LeRobot, ACT, SmolVLA, π_0, LoRA

- Brought up a leader-follower SO-101 arm pair in LeRobot; calibrating across arms so a policy trained on one runs on the other, with a demo protocol of fixed cameras, one grasp strategy, and replay checks.
- Next: on one language-conditioned task, benchmark a from-scratch ACT baseline against LoRA fine-tuned VLAs (SmolVLA, π_0) on a 16 GB GPU, using asynchronous action-chunk inference.

Sonar-to-Vision Transformer (independent research, Dec 2025 - present) — PyTorch, ROS 2, TensorRT

- Developing a Transformer that maps imaging-sonar returns to vision-style features so AcTag fiducials stay detectable in murky water; deploying via TensorRT (FP16) on a Jetson Orin Nano into a ROS 2 EKF stack.

SLAM Path-Planning Robot — ROS 2, Python/C++, PyTorch, TensorRT, LiDAR/IMU

- Fused LiDAR, camera, and IMU data through a NumPy EKF into a semantic BEV occupancy grid that feeds A* planning; detector and segmenter models run on a Jetson Orin Nano as TensorRT (FP16) ROS 2 nodes.

Technical Skills

Programming: Python, C/C++, SQL, JavaScript, R
Robotics: ROS 2 Humble, Nav2, SLAM, EKF fusion (robot_localization, tf2), PID control, A* planning
Perception / ML: PyTorch, YOLO, OpenCV, ONNX, TensorRT, llama.cpp/GGUF, Whisper, LeRobot, scikit-learn
Currently learning: Isaac Sim/Lab, RL (PPO), sim-to-real, imitation learning (ACT), VLA fine-tuning (LoRA)
Hardware: NVIDIA Jetson Orin NX/Nano, LiDAR, IMU, RGB-D cameras, ReSpeaker mic array, SO-101 arm
Tools: Linux, Docker, Git, PostgreSQL, Grafana, FastAPI, Flask | **Spoken:** English (native), German (conversational)

Activities & Affiliations

Robotics@Maryland, Qubo AUV (Perception/CV), Aug 2025 - Jan 2026: YOLO and ROS 2 vision pipeline on a Jetson Orin Nano, tuned for stable frame rate and low latency | FIRST Team 4514 (Programming Lead) | UMD BattleBots (Leatherback) | Virginia Tech TREC Lab (ESCHER humanoid wiring)